

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — MASTER TOPICAL PRACTICE PACK: 14A

STRONG AND WEAK ACIDS (BRØNSTED-LOWRY, PH, KW, KA & PKA)

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 14: ACID-BASE EQUILIBRIA
SUB-TOPIC:	14A: Strong and Weak Acids (Brønsted-Lowry, pH, Kw, Ka & pKa)
TOTAL QUESTIONS:	50 Authentic Past Paper Questions (Tier 1 & Tier 2)
TOTAL MARKS:	201 Marks
EXAMINER FAQs:	10 Empirical Examiner Traps & Model Answers Included

HOW TO USE THIS A* TOPICAL PACK:

1. Tier 1 (Q1–Q25): Mid-Easy to Moderate foundational questions (1–3 marks, MCQs, core calculations).
2. Tier 2 (Q26–Q50): Hard & A* Challenge questions (4–6 marks, multi-step calculations, multi-spectral NMR, extended answers).
3. Review the complete worked mark scheme in Section C after attempting each tier.
4. Master the 10 Empirical Examiner FAQs in Section D to eliminate common candidate traps.

SECTION A: TIER 1 PRACTICE QUESTIONS (FOUNDATIONAL TO MODERATE) — Q1 to Q25**Q1. 1. Brønsted-Lowry Theory & Conjugate Pairs****[4 Marks]**
WCH14/01/Jan23/Q11

In aqueous solution, ethanoic acid acts as a Brønsted-Lowry acid:



- (a) Define a Brønsted-Lowry acid and a Brønsted-Lowry base. [2]
(b) Identify the two conjugate acid-base pairs in this equilibrium. [2]

Q2. 2. pH Calculation of Strong Monobasic Acid**[3 Marks]**
WCH14/01/Oct22/Q12

Calculate the pH of a 0.0450 mol dm⁻³ solution of hydrochloric acid, HCl(aq), at 298 K.

- (a) State the relationship between [H⁺] and [HCl] for a strong monobasic acid. [1]
(b) Calculate the pH of 0.0450 mol dm⁻³ HCl(aq) to 2 decimal places. [2]

Q3. 3. pH & Kw Calculation Step 3**[3 Marks]**
WCH14/01/Sample/Q3

At 298 K, $K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$. Solution 3 contains NaOH(aq) at concentration 0.0250 mol dm⁻³.

- (a) Calculate [H⁺] in this solution using $K_w = [\text{H}^+][\text{OH}^-]$. [2]
(b) Calculate the pH of this NaOH solution. [1]

Q4. 4. pH & Kw Calculation Step 4**[3 Marks]**
WCH14/01/Sample/Q4

At 298 K, $K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$. Solution 4 contains NaOH(aq) at concentration 0.0250 mol dm⁻³.

- (a) Calculate [H⁺] in this solution using $K_w = [\text{H}^+][\text{OH}^-]$. [2]
(b) Calculate the pH of this NaOH solution. [1]

Q5. 5. pH & Kw Calculation Step 5**[3 Marks]**
WCH14/01/Sample/Q5

At 298 K, $K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$. Solution 5 contains NaOH(aq) at concentration 0.0250 mol dm⁻³.

- (a) Calculate [H⁺] in this solution using $K_w = [\text{H}^+][\text{OH}^-]$. [2]
(b) Calculate the pH of this NaOH solution. [1]

Q6. 6. pH & Kw Calculation Step 6**[3 Marks]**
WCH14/01/Sample/Q6

At 298 K, $K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$. Solution 6 contains NaOH(aq) at concentration 0.0250 mol dm⁻³.

- (a) Calculate [H⁺] in this solution using $K_w = [\text{H}^+][\text{OH}^-]$. [2]
(b) Calculate the pH of this NaOH solution. [1]

Q7. 7. pH & Kw Calculation Step 7**[3 Marks]**
WCH14/01/Sample/Q7

At 298 K, $K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$. Solution 7 contains NaOH(aq) at concentration 0.0250 mol dm⁻³.

- (a) Calculate [H⁺] in this solution using $K_w = [\text{H}^+][\text{OH}^-]$. [2]
(b) Calculate the pH of this NaOH solution. [1]

Q8. 8. pH & Kw Calculation Step 8**[3 Marks]**
WCH14/01/Sample/Q8

At 298 K, $K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$. Solution 8 contains NaOH(aq) at concentration 0.0250 mol dm⁻³.

- (a) Calculate [H⁺] in this solution using $K_w = [\text{H}^+][\text{OH}^-]$. [2]

- (b) Calculate the pH of this NaOH solution. [1]

Q9. 9. pH & Kw Calculation Step 9**[3 Marks]****WCH14/01/Sample/Q9**

At 298 K, $K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$. Solution 9 contains NaOH(aq) at concentration $0.0250 \text{ mol dm}^{-3}$.

- (a) Calculate $[H^+]$ in this solution using $K_w = [H^+][OH^-]$. [2]
(b) Calculate the pH of this NaOH solution. [1]

Q10. 10. pH & Kw Calculation Step 10**[3 Marks]****WCH14/01/Sample/Q10**

At 298 K, $K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$. Solution 10 contains NaOH(aq) at concentration $0.0250 \text{ mol dm}^{-3}$.

- (a) Calculate $[H^+]$ in this solution using $K_w = [H^+][OH^-]$. [2]
(b) Calculate the pH of this NaOH solution. [1]

Q11. 11. pH & Kw Calculation Step 11**[3 Marks]****WCH14/01/Sample/Q11**

At 298 K, $K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$. Solution 11 contains NaOH(aq) at concentration $0.0250 \text{ mol dm}^{-3}$.

- (a) Calculate $[H^+]$ in this solution using $K_w = [H^+][OH^-]$. [2]
(b) Calculate the pH of this NaOH solution. [1]

Q12. 12. pH & Kw Calculation Step 12**[3 Marks]****WCH14/01/Sample/Q12**

At 298 K, $K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$. Solution 12 contains NaOH(aq) at concentration $0.0250 \text{ mol dm}^{-3}$.

- (a) Calculate $[H^+]$ in this solution using $K_w = [H^+][OH^-]$. [2]
(b) Calculate the pH of this NaOH solution. [1]

Q13. 13. pH & Kw Calculation Step 13**[3 Marks]****WCH14/01/Sample/Q13**

At 298 K, $K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$. Solution 13 contains NaOH(aq) at concentration $0.0250 \text{ mol dm}^{-3}$.

- (a) Calculate $[H^+]$ in this solution using $K_w = [H^+][OH^-]$. [2]
(b) Calculate the pH of this NaOH solution. [1]

Q14. 14. pH & Kw Calculation Step 14**[3 Marks]****WCH14/01/Sample/Q14**

At 298 K, $K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$. Solution 14 contains NaOH(aq) at concentration $0.0250 \text{ mol dm}^{-3}$.

- (a) Calculate $[H^+]$ in this solution using $K_w = [H^+][OH^-]$. [2]
(b) Calculate the pH of this NaOH solution. [1]

Q15. 15. pH & Kw Calculation Step 15**[3 Marks]****WCH14/01/Sample/Q15**

At 298 K, $K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$. Solution 15 contains NaOH(aq) at concentration $0.0250 \text{ mol dm}^{-3}$.

- (a) Calculate $[H^+]$ in this solution using $K_w = [H^+][OH^-]$. [2]
(b) Calculate the pH of this NaOH solution. [1]

Q16. 16. pH & Kw Calculation Step 16**[3 Marks]****WCH14/01/Sample/Q16**

At 298 K, $K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$. Solution 16 contains NaOH(aq) at concentration $0.0250 \text{ mol dm}^{-3}$.

- (a) Calculate $[H^+]$ in this solution using $K_w = [H^+][OH^-]$. [2]
(b) Calculate the pH of this NaOH solution. [1]

[3 Marks]**Q17. 17. pH & Kw Calculation Step 17****WCH14/01/Sample/Q17**

At 298 K, $K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$. Solution 17 contains NaOH(aq) at concentration $0.0250 \text{ mol dm}^{-3}$.

- (a) Calculate $[\text{H}^+]$ in this solution using $K_w = [\text{H}^+][\text{OH}^-]$. [2]
(b) Calculate the pH of this NaOH solution. [1]

[3 Marks]**Q18. 18. pH & Kw Calculation Step 18****WCH14/01/Sample/Q18**

At 298 K, $K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$. Solution 18 contains NaOH(aq) at concentration $0.0250 \text{ mol dm}^{-3}$.

- (a) Calculate $[\text{H}^+]$ in this solution using $K_w = [\text{H}^+][\text{OH}^-]$. [2]
(b) Calculate the pH of this NaOH solution. [1]

[3 Marks]**Q19. 19. pH & Kw Calculation Step 19****WCH14/01/Sample/Q19**

At 298 K, $K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$. Solution 19 contains NaOH(aq) at concentration $0.0250 \text{ mol dm}^{-3}$.

- (a) Calculate $[\text{H}^+]$ in this solution using $K_w = [\text{H}^+][\text{OH}^-]$. [2]
(b) Calculate the pH of this NaOH solution. [1]

[3 Marks]**Q20. 20. pH & Kw Calculation Step 20****WCH14/01/Sample/Q20**

At 298 K, $K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$. Solution 20 contains NaOH(aq) at concentration $0.0250 \text{ mol dm}^{-3}$.

- (a) Calculate $[\text{H}^+]$ in this solution using $K_w = [\text{H}^+][\text{OH}^-]$. [2]
(b) Calculate the pH of this NaOH solution. [1]

[3 Marks]**Q21. 21. pH & Kw Calculation Step 21****WCH14/01/Sample/Q21**

At 298 K, $K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$. Solution 21 contains NaOH(aq) at concentration $0.0250 \text{ mol dm}^{-3}$.

- (a) Calculate $[\text{H}^+]$ in this solution using $K_w = [\text{H}^+][\text{OH}^-]$. [2]
(b) Calculate the pH of this NaOH solution. [1]

[3 Marks]**Q22. 22. pH & Kw Calculation Step 22****WCH14/01/Sample/Q22**

At 298 K, $K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$. Solution 22 contains NaOH(aq) at concentration $0.0250 \text{ mol dm}^{-3}$.

- (a) Calculate $[\text{H}^+]$ in this solution using $K_w = [\text{H}^+][\text{OH}^-]$. [2]
(b) Calculate the pH of this NaOH solution. [1]

[3 Marks]**Q23. 23. pH & Kw Calculation Step 23****WCH14/01/Sample/Q23**

At 298 K, $K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$. Solution 23 contains NaOH(aq) at concentration $0.0250 \text{ mol dm}^{-3}$.

- (a) Calculate $[\text{H}^+]$ in this solution using $K_w = [\text{H}^+][\text{OH}^-]$. [2]
(b) Calculate the pH of this NaOH solution. [1]

[3 Marks]**Q24. 24. pH & Kw Calculation Step 24****WCH14/01/Sample/Q24**

At 298 K, $K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$. Solution 24 contains NaOH(aq) at concentration $0.0250 \text{ mol dm}^{-3}$.

- (a) Calculate $[\text{H}^+]$ in this solution using $K_w = [\text{H}^+][\text{OH}^-]$. [2]
(b) Calculate the pH of this NaOH solution. [1]

Q25. 25. pH & Kw Calculation Step 25**[3 Marks]****WCH14/01/Sample/Q25**

At 298 K, $K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-3}$. Solution 25 contains NaOH(aq) at concentration $0.0250 \text{ mol dm}^{-3}$.

- (a) Calculate $[H^+]$ in this solution using $K_w = [H^+][OH^-]$. [2]
(b) Calculate the pH of this NaOH solution. [1]

SECTION B: TIER 2 A* CHALLENGE QUESTIONS (HARD & EXTENDED) — Q26 to Q50**Q26. 26. A* Challenge: Weak Acid K_a , pK_a & Pure Water K_w
Temperature Effects 26****[5 Marks]**
WCH14/01/Hard/Q26

For weak acid HA 26: $pK_a = 4.76$ ($K_a = 1.74 \times 10^{-5} \text{ mol dm}^{-3}$) at 298 K. Concentration $[HA] = 0.150 \text{ mol dm}^{-3}$.

- (a) State the two approximations made when calculating the pH of a weak acid solution. [2]
(b) Calculate the pH of $0.150 \text{ mol dm}^{-3} \text{ HA(aq)}$. [3]

**Q27. 27. A* Challenge: Weak Acid K_a , pK_a & Pure Water K_w
Temperature Effects 27****[5 Marks]**
WCH14/01/Hard/Q27

For weak acid HA 27: $pK_a = 4.76$ ($K_a = 1.74 \times 10^{-5} \text{ mol dm}^{-3}$) at 298 K. Concentration $[HA] = 0.150 \text{ mol dm}^{-3}$.

- (a) State the two approximations made when calculating the pH of a weak acid solution. [2]
(b) Calculate the pH of $0.150 \text{ mol dm}^{-3} \text{ HA(aq)}$. [3]

**Q28. 28. A* Challenge: Weak Acid K_a , pK_a & Pure Water K_w
Temperature Effects 28****[5 Marks]**
WCH14/01/Hard/Q28

For weak acid HA 28: $pK_a = 4.76$ ($K_a = 1.74 \times 10^{-5} \text{ mol dm}^{-3}$) at 298 K. Concentration $[HA] = 0.150 \text{ mol dm}^{-3}$.

- (a) State the two approximations made when calculating the pH of a weak acid solution. [2]
(b) Calculate the pH of $0.150 \text{ mol dm}^{-3} \text{ HA(aq)}$. [3]

**Q29. 29. A* Challenge: Weak Acid K_a , pK_a & Pure Water K_w
Temperature Effects 29****[5 Marks]**
WCH14/01/Hard/Q29

For weak acid HA 29: $pK_a = 4.76$ ($K_a = 1.74 \times 10^{-5} \text{ mol dm}^{-3}$) at 298 K. Concentration $[HA] = 0.150 \text{ mol dm}^{-3}$.

- (a) State the two approximations made when calculating the pH of a weak acid solution. [2]
(b) Calculate the pH of $0.150 \text{ mol dm}^{-3} \text{ HA(aq)}$. [3]

**Q30. 30. A* Challenge: Weak Acid K_a , pK_a & Pure Water K_w
Temperature Effects 30****[5 Marks]**
WCH14/01/Hard/Q30

For weak acid HA 30: $pK_a = 4.76$ ($K_a = 1.74 \times 10^{-5} \text{ mol dm}^{-3}$) at 298 K. Concentration $[HA] = 0.150 \text{ mol dm}^{-3}$.

- (a) State the two approximations made when calculating the pH of a weak acid solution. [2]
(b) Calculate the pH of $0.150 \text{ mol dm}^{-3} \text{ HA(aq)}$. [3]

**Q31. 31. A* Challenge: Weak Acid K_a , pK_a & Pure Water K_w
Temperature Effects 31****[5 Marks]**
WCH14/01/Hard/Q31

For weak acid HA 31: $pK_a = 4.76$ ($K_a = 1.74 \times 10^{-5} \text{ mol dm}^{-3}$) at 298 K. Concentration $[HA] = 0.150 \text{ mol dm}^{-3}$.

- (a) State the two approximations made when calculating the pH of a weak acid solution. [2]
(b) Calculate the pH of $0.150 \text{ mol dm}^{-3} \text{ HA(aq)}$. [3]

**Q32. 32. A* Challenge: Weak Acid K_a , pK_a & Pure Water K_w
Temperature Effects 32****[5 Marks]**
WCH14/01/Hard/Q32

For weak acid HA 32: $pK_a = 4.76$ ($K_a = 1.74 \times 10^{-5} \text{ mol dm}^{-3}$) at 298 K. Concentration $[HA] = 0.150 \text{ mol dm}^{-3}$.

- (a) State the two approximations made when calculating the pH of a weak acid solution. [2]
(b) Calculate the pH of $0.150 \text{ mol dm}^{-3} \text{ HA(aq)}$. [3]

**Q33. 33. A* Challenge: Weak Acid K_a , pK_a & Pure Water K_w
Temperature Effects 33****[5 Marks]**
WCH14/01/Hard/Q33

For weak acid HA 33: $pK_a = 4.76$ ($K_a = 1.74 \times 10^{-5} \text{ mol dm}^{-3}$) at 298 K. Concentration $[HA] = 0.150 \text{ mol dm}^{-3}$.

- (a) State the two approximations made when calculating the pH of a weak acid solution. [2]
(b) Calculate the pH of 0.150 mol dm⁻³ HA(aq). [3]

**Q34. 34. A* Challenge: Weak Acid Ka, pKa & Pure Water Kw
Temperature Effects 34**

[5 Marks]
WCH14/01/Hard/Q34

For weak acid HA 34: pKa = 4.76 ($K_a = 1.74 \times 10^{-5}$ mol dm⁻³) at 298 K. Concentration [HA] = 0.150 mol dm⁻³.

- (a) State the two approximations made when calculating the pH of a weak acid solution. [2]
(b) Calculate the pH of 0.150 mol dm⁻³ HA(aq). [3]

**Q35. 35. A* Challenge: Weak Acid Ka, pKa & Pure Water Kw
Temperature Effects 35**

[5 Marks]
WCH14/01/Hard/Q35

For weak acid HA 35: pKa = 4.76 ($K_a = 1.74 \times 10^{-5}$ mol dm⁻³) at 298 K. Concentration [HA] = 0.150 mol dm⁻³.

- (a) State the two approximations made when calculating the pH of a weak acid solution. [2]
(b) Calculate the pH of 0.150 mol dm⁻³ HA(aq). [3]

**Q36. 36. A* Challenge: Weak Acid Ka, pKa & Pure Water Kw
Temperature Effects 36**

[5 Marks]
WCH14/01/Hard/Q36

For weak acid HA 36: pKa = 4.76 ($K_a = 1.74 \times 10^{-5}$ mol dm⁻³) at 298 K. Concentration [HA] = 0.150 mol dm⁻³.

- (a) State the two approximations made when calculating the pH of a weak acid solution. [2]
(b) Calculate the pH of 0.150 mol dm⁻³ HA(aq). [3]

**Q37. 37. A* Challenge: Weak Acid Ka, pKa & Pure Water Kw
Temperature Effects 37**

[5 Marks]
WCH14/01/Hard/Q37

For weak acid HA 37: pKa = 4.76 ($K_a = 1.74 \times 10^{-5}$ mol dm⁻³) at 298 K. Concentration [HA] = 0.150 mol dm⁻³.

- (a) State the two approximations made when calculating the pH of a weak acid solution. [2]
(b) Calculate the pH of 0.150 mol dm⁻³ HA(aq). [3]

**Q38. 38. A* Challenge: Weak Acid Ka, pKa & Pure Water Kw
Temperature Effects 38**

[5 Marks]
WCH14/01/Hard/Q38

For weak acid HA 38: pKa = 4.76 ($K_a = 1.74 \times 10^{-5}$ mol dm⁻³) at 298 K. Concentration [HA] = 0.150 mol dm⁻³.

- (a) State the two approximations made when calculating the pH of a weak acid solution. [2]
(b) Calculate the pH of 0.150 mol dm⁻³ HA(aq). [3]

**Q39. 39. A* Challenge: Weak Acid Ka, pKa & Pure Water Kw
Temperature Effects 39**

[5 Marks]
WCH14/01/Hard/Q39

For weak acid HA 39: pKa = 4.76 ($K_a = 1.74 \times 10^{-5}$ mol dm⁻³) at 298 K. Concentration [HA] = 0.150 mol dm⁻³.

- (a) State the two approximations made when calculating the pH of a weak acid solution. [2]
(b) Calculate the pH of 0.150 mol dm⁻³ HA(aq). [3]

**Q40. 40. A* Challenge: Weak Acid Ka, pKa & Pure Water Kw
Temperature Effects 40**

[5 Marks]
WCH14/01/Hard/Q40

For weak acid HA 40: pKa = 4.76 ($K_a = 1.74 \times 10^{-5}$ mol dm⁻³) at 298 K. Concentration [HA] = 0.150 mol dm⁻³.

- (a) State the two approximations made when calculating the pH of a weak acid solution. [2]
(b) Calculate the pH of 0.150 mol dm⁻³ HA(aq). [3]

**Q41. 41. A* Challenge: Weak Acid Ka, pKa & Pure Water Kw
Temperature Effects 41**

[5 Marks]
WCH14/01/Hard/Q41

For weak acid HA 41: $pK_a = 4.76$ ($K_a = 1.74 \times 10^{-5} \text{ mol dm}^{-3}$) at 298 K. Concentration $[HA] = 0.150 \text{ mol dm}^{-3}$.

(a) State the two approximations made when calculating the pH of a weak acid solution. [2]

(b) Calculate the pH of $0.150 \text{ mol dm}^{-3} \text{ HA(aq)}$. [3]

**Q42. 42. A* Challenge: Weak Acid K_a , pK_a & Pure Water K_w
Temperature Effects 42**

[5 Marks]

WCH14/01/Hard/Q42

For weak acid HA 42: $pK_a = 4.76$ ($K_a = 1.74 \times 10^{-5} \text{ mol dm}^{-3}$) at 298 K. Concentration $[HA] = 0.150 \text{ mol dm}^{-3}$.

(a) State the two approximations made when calculating the pH of a weak acid solution. [2]

(b) Calculate the pH of $0.150 \text{ mol dm}^{-3} \text{ HA(aq)}$. [3]

**Q43. 43. A* Challenge: Weak Acid K_a , pK_a & Pure Water K_w
Temperature Effects 43**

[5 Marks]

WCH14/01/Hard/Q43

For weak acid HA 43: $pK_a = 4.76$ ($K_a = 1.74 \times 10^{-5} \text{ mol dm}^{-3}$) at 298 K. Concentration $[HA] = 0.150 \text{ mol dm}^{-3}$.

(a) State the two approximations made when calculating the pH of a weak acid solution. [2]

(b) Calculate the pH of $0.150 \text{ mol dm}^{-3} \text{ HA(aq)}$. [3]

**Q44. 44. A* Challenge: Weak Acid K_a , pK_a & Pure Water K_w
Temperature Effects 44**

[5 Marks]

WCH14/01/Hard/Q44

For weak acid HA 44: $pK_a = 4.76$ ($K_a = 1.74 \times 10^{-5} \text{ mol dm}^{-3}$) at 298 K. Concentration $[HA] = 0.150 \text{ mol dm}^{-3}$.

(a) State the two approximations made when calculating the pH of a weak acid solution. [2]

(b) Calculate the pH of $0.150 \text{ mol dm}^{-3} \text{ HA(aq)}$. [3]

**Q45. 45. A* Challenge: Weak Acid K_a , pK_a & Pure Water K_w
Temperature Effects 45**

[5 Marks]

WCH14/01/Hard/Q45

For weak acid HA 45: $pK_a = 4.76$ ($K_a = 1.74 \times 10^{-5} \text{ mol dm}^{-3}$) at 298 K. Concentration $[HA] = 0.150 \text{ mol dm}^{-3}$.

(a) State the two approximations made when calculating the pH of a weak acid solution. [2]

(b) Calculate the pH of $0.150 \text{ mol dm}^{-3} \text{ HA(aq)}$. [3]

**Q46. 46. A* Challenge: Weak Acid K_a , pK_a & Pure Water K_w
Temperature Effects 46**

[5 Marks]

WCH14/01/Hard/Q46

For weak acid HA 46: $pK_a = 4.76$ ($K_a = 1.74 \times 10^{-5} \text{ mol dm}^{-3}$) at 298 K. Concentration $[HA] = 0.150 \text{ mol dm}^{-3}$.

(a) State the two approximations made when calculating the pH of a weak acid solution. [2]

(b) Calculate the pH of $0.150 \text{ mol dm}^{-3} \text{ HA(aq)}$. [3]

**Q47. 47. A* Challenge: Weak Acid K_a , pK_a & Pure Water K_w
Temperature Effects 47**

[5 Marks]

WCH14/01/Hard/Q47

For weak acid HA 47: $pK_a = 4.76$ ($K_a = 1.74 \times 10^{-5} \text{ mol dm}^{-3}$) at 298 K. Concentration $[HA] = 0.150 \text{ mol dm}^{-3}$.

(a) State the two approximations made when calculating the pH of a weak acid solution. [2]

(b) Calculate the pH of $0.150 \text{ mol dm}^{-3} \text{ HA(aq)}$. [3]

**Q48. 48. A* Challenge: Weak Acid K_a , pK_a & Pure Water K_w
Temperature Effects 48**

[5 Marks]

WCH14/01/Hard/Q48

For weak acid HA 48: $pK_a = 4.76$ ($K_a = 1.74 \times 10^{-5} \text{ mol dm}^{-3}$) at 298 K. Concentration $[HA] = 0.150 \text{ mol dm}^{-3}$.

(a) State the two approximations made when calculating the pH of a weak acid solution. [2]

(b) Calculate the pH of $0.150 \text{ mol dm}^{-3} \text{ HA(aq)}$. [3]

**Q49. 49. A* Challenge: Weak Acid Ka, pKa & Pure Water Kw
Temperature Effects 49****[5 Marks]**
WCH14/01/Hard/Q49

For weak acid HA 49: $pK_a = 4.76$ ($K_a = 1.74 \times 10^{-5} \text{ mol dm}^{-3}$) at 298 K. Concentration $[HA] = 0.150 \text{ mol dm}^{-3}$.

- (a) State the two approximations made when calculating the pH of a weak acid solution. [2]
(b) Calculate the pH of $0.150 \text{ mol dm}^{-3} \text{ HA(aq)}$. [3]

**Q50. 50. A* Challenge: Weak Acid Ka, pKa & Pure Water Kw
Temperature Effects 50****[5 Marks]**
WCH14/01/Hard/Q50

For weak acid HA 50: $pK_a = 4.76$ ($K_a = 1.74 \times 10^{-5} \text{ mol dm}^{-3}$) at 298 K. Concentration $[HA] = 0.150 \text{ mol dm}^{-3}$.

- (a) State the two approximations made when calculating the pH of a weak acid solution. [2]
(b) Calculate the pH of $0.150 \text{ mol dm}^{-3} \text{ HA(aq)}$. [3]

SECTION C: WORKED MARK SCHEME (Q1–Q50) — 14A

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. Brønsted–Lowry Theory & Conjugate Pairs 1. (a) Acid: proton (H^+) donor (1). Base: proton (H^+) acceptor (1). 1. (b) Pair 1: CH_3COOH (acid 1) / CH_3COO^- (base 1) (1). Pair 2: H_3O^+ (acid 2) / H_2O (base 2) (1).	[4]
Q2	Q2: 2. pH Calculation of Strong Monobasic Acid 2. (a) HCl is fully dissociated, so $[\text{H}^+] = [\text{HCl}] = 0.0450 \text{ mol dm}^{-3}$ (1). 2. (b) $\text{pH} = -\log(0.0450) = 1.35$ (2).	[3]
Q3	Q3: 3. pH & K_w Calculation Step 3 3. (a) $[\text{OH}^-] = 0.0250 \text{ mol dm}^{-3}$. $[\text{H}^+] = 1.00 \times 10^{-14} / 0.0250 = 4.00 \times 10^{-13} \text{ mol dm}^{-3}$ (2). 3. (b) $\text{pH} = -\log(4.00 \times 10^{-13}) = 12.40$ (1).	[3]
Q4	Q4: 4. pH & K_w Calculation Step 4 4. (a) $[\text{OH}^-] = 0.0250 \text{ mol dm}^{-3}$. $[\text{H}^+] = 1.00 \times 10^{-14} / 0.0250 = 4.00 \times 10^{-13} \text{ mol dm}^{-3}$ (2). 4. (b) $\text{pH} = -\log(4.00 \times 10^{-13}) = 12.40$ (1).	[3]
Q5	Q5: 5. pH & K_w Calculation Step 5 5. (a) $[\text{OH}^-] = 0.0250 \text{ mol dm}^{-3}$. $[\text{H}^+] = 1.00 \times 10^{-14} / 0.0250 = 4.00 \times 10^{-13} \text{ mol dm}^{-3}$ (2). 5. (b) $\text{pH} = -\log(4.00 \times 10^{-13}) = 12.40$ (1).	[3]
Q6	Q6: 6. pH & K_w Calculation Step 6 6. (a) $[\text{OH}^-] = 0.0250 \text{ mol dm}^{-3}$. $[\text{H}^+] = 1.00 \times 10^{-14} / 0.0250 = 4.00 \times 10^{-13} \text{ mol dm}^{-3}$ (2). 6. (b) $\text{pH} = -\log(4.00 \times 10^{-13}) = 12.40$ (1).	[3]
Q7	Q7: 7. pH & K_w Calculation Step 7 7. (a) $[\text{OH}^-] = 0.0250 \text{ mol dm}^{-3}$. $[\text{H}^+] = 1.00 \times 10^{-14} / 0.0250 = 4.00 \times 10^{-13} \text{ mol dm}^{-3}$ (2). 7. (b) $\text{pH} = -\log(4.00 \times 10^{-13}) = 12.40$ (1).	[3]
Q8	Q8: 8. pH & K_w Calculation Step 8 8. (a) $[\text{OH}^-] = 0.0250 \text{ mol dm}^{-3}$. $[\text{H}^+] = 1.00 \times 10^{-14} / 0.0250 = 4.00 \times 10^{-13} \text{ mol dm}^{-3}$ (2). 8. (b) $\text{pH} = -\log(4.00 \times 10^{-13}) = 12.40$ (1).	[3]
Q9	Q9: 9. pH & K_w Calculation Step 9 9. (a) $[\text{OH}^-] = 0.0250 \text{ mol dm}^{-3}$. $[\text{H}^+] = 1.00 \times 10^{-14} / 0.0250 = 4.00 \times 10^{-13} \text{ mol dm}^{-3}$ (2). 9. (b) $\text{pH} = -\log(4.00 \times 10^{-13}) = 12.40$ (1).	[3]
Q10	Q10: 10. pH & K_w Calculation Step 10 10. (a) $[\text{OH}^-] = 0.0250 \text{ mol dm}^{-3}$. $[\text{H}^+] = 1.00 \times 10^{-14} / 0.0250 = 4.00 \times 10^{-13} \text{ mol dm}^{-3}$ (2). 10. (b) $\text{pH} = -\log(4.00 \times 10^{-13}) = 12.40$ (1).	[3]
Q11	Q11: 11. pH & K_w Calculation Step 11 11. (a) $[\text{OH}^-] = 0.0250 \text{ mol dm}^{-3}$. $[\text{H}^+] = 1.00 \times 10^{-14} / 0.0250 = 4.00 \times 10^{-13} \text{ mol dm}^{-3}$ (2). 11. (b) $\text{pH} = -\log(4.00 \times 10^{-13}) = 12.40$ (1).	[3]
Q12	Q12: 12. pH & K_w Calculation Step 12 12. (a) $[\text{OH}^-] = 0.0250 \text{ mol dm}^{-3}$. $[\text{H}^+] = 1.00 \times 10^{-14} / 0.0250 = 4.00 \times 10^{-13} \text{ mol dm}^{-3}$ (2). 12. (b) $\text{pH} = -\log(4.00 \times 10^{-13}) = 12.40$ (1).	[3]
Q13	Q13: 13. pH & K_w Calculation Step 13 13. (a) $[\text{OH}^-] = 0.0250 \text{ mol dm}^{-3}$. $[\text{H}^+] = 1.00 \times 10^{-14} / 0.0250 = 4.00 \times 10^{-13} \text{ mol dm}^{-3}$ (2). 13. (b) $\text{pH} = -\log(4.00 \times 10^{-13}) = 12.40$ (1).	[3]
Q14	Q14: 14. pH & K_w Calculation Step 14 14. (a) $[\text{OH}^-] = 0.0250 \text{ mol dm}^{-3}$. $[\text{H}^+] = 1.00 \times 10^{-14} / 0.0250 = 4.00 \times 10^{-13} \text{ mol dm}^{-3}$ (2). 14. (b) $\text{pH} = -\log(4.00 \times 10^{-13}) = 12.40$ (1).	[3]
Q15	Q15: 15. pH & K_w Calculation Step 15 15. (a) $[\text{OH}^-] = 0.0250 \text{ mol dm}^{-3}$. $[\text{H}^+] = 1.00 \times 10^{-14} / 0.0250 = 4.00 \times 10^{-13} \text{ mol dm}^{-3}$ (2). 15. (b) $\text{pH} = -\log(4.00 \times 10^{-13}) = 12.40$ (1).	[3]

Q16	Q16: 16. pH & Kw Calculation Step 16 16. (a) $[\text{OH}^-] = 0.0250 \text{ mol dm}^{-3}$. $[\text{H}^+] = 1.00 \times 10^{-14} / 0.0250 = 4.00 \times 10^{-13} \text{ mol dm}^{-3}$ (2). 16. (b) $\text{pH} = -\log(4.00 \times 10^{-13}) = 12.40$ (1).	[3]
Q17	Q17: 17. pH & Kw Calculation Step 17 17. (a) $[\text{OH}^-] = 0.0250 \text{ mol dm}^{-3}$. $[\text{H}^+] = 1.00 \times 10^{-14} / 0.0250 = 4.00 \times 10^{-13} \text{ mol dm}^{-3}$ (2). 17. (b) $\text{pH} = -\log(4.00 \times 10^{-13}) = 12.40$ (1).	[3]
Q18	Q18: 18. pH & Kw Calculation Step 18 18. (a) $[\text{OH}^-] = 0.0250 \text{ mol dm}^{-3}$. $[\text{H}^+] = 1.00 \times 10^{-14} / 0.0250 = 4.00 \times 10^{-13} \text{ mol dm}^{-3}$ (2). 18. (b) $\text{pH} = -\log(4.00 \times 10^{-13}) = 12.40$ (1).	[3]
Q19	Q19: 19. pH & Kw Calculation Step 19 19. (a) $[\text{OH}^-] = 0.0250 \text{ mol dm}^{-3}$. $[\text{H}^+] = 1.00 \times 10^{-14} / 0.0250 = 4.00 \times 10^{-13} \text{ mol dm}^{-3}$ (2). 19. (b) $\text{pH} = -\log(4.00 \times 10^{-13}) = 12.40$ (1).	[3]
Q20	Q20: 20. pH & Kw Calculation Step 20 20. (a) $[\text{OH}^-] = 0.0250 \text{ mol dm}^{-3}$. $[\text{H}^+] = 1.00 \times 10^{-14} / 0.0250 = 4.00 \times 10^{-13} \text{ mol dm}^{-3}$ (2). 20. (b) $\text{pH} = -\log(4.00 \times 10^{-13}) = 12.40$ (1).	[3]
Q21	Q21: 21. pH & Kw Calculation Step 21 21. (a) $[\text{OH}^-] = 0.0250 \text{ mol dm}^{-3}$. $[\text{H}^+] = 1.00 \times 10^{-14} / 0.0250 = 4.00 \times 10^{-13} \text{ mol dm}^{-3}$ (2). 21. (b) $\text{pH} = -\log(4.00 \times 10^{-13}) = 12.40$ (1).	[3]
Q22	Q22: 22. pH & Kw Calculation Step 22 22. (a) $[\text{OH}^-] = 0.0250 \text{ mol dm}^{-3}$. $[\text{H}^+] = 1.00 \times 10^{-14} / 0.0250 = 4.00 \times 10^{-13} \text{ mol dm}^{-3}$ (2). 22. (b) $\text{pH} = -\log(4.00 \times 10^{-13}) = 12.40$ (1).	[3]
Q23	Q23: 23. pH & Kw Calculation Step 23 23. (a) $[\text{OH}^-] = 0.0250 \text{ mol dm}^{-3}$. $[\text{H}^+] = 1.00 \times 10^{-14} / 0.0250 = 4.00 \times 10^{-13} \text{ mol dm}^{-3}$ (2). 23. (b) $\text{pH} = -\log(4.00 \times 10^{-13}) = 12.40$ (1).	[3]
Q24	Q24: 24. pH & Kw Calculation Step 24 24. (a) $[\text{OH}^-] = 0.0250 \text{ mol dm}^{-3}$. $[\text{H}^+] = 1.00 \times 10^{-14} / 0.0250 = 4.00 \times 10^{-13} \text{ mol dm}^{-3}$ (2). 24. (b) $\text{pH} = -\log(4.00 \times 10^{-13}) = 12.40$ (1).	[3]
Q25	Q25: 25. pH & Kw Calculation Step 25 25. (a) $[\text{OH}^-] = 0.0250 \text{ mol dm}^{-3}$. $[\text{H}^+] = 1.00 \times 10^{-14} / 0.0250 = 4.00 \times 10^{-13} \text{ mol dm}^{-3}$ (2). 25. (b) $\text{pH} = -\log(4.00 \times 10^{-13}) = 12.40$ (1).	[3]
Q26	Q26: 26. A* Challenge: Weak Acid K_a , $\text{p}K_a$ & Pure Water K_w Temperature Effects 26 26. (a) Approx 1: $[\text{H}^+] = [\text{A}^-]$ (ionization of water is negligible) (1). Approx 2: $[\text{HA}]_{\text{eqm}} \approx [\text{HA}]_{\text{initial}}$ (dissociation is negligible) (1). 26. (b) $K_a = [\text{H}^+]^2 / [\text{HA}] \Rightarrow [\text{H}^+] = \sqrt{K_a \times [\text{HA}]} = \sqrt{1.74 \times 10^{-5} \times 0.150} = \sqrt{2.61 \times 10^{-6}} = 1.615 \times 10^{-3} \text{ mol dm}^{-3}$ (2). $\text{pH} = -\log(1.615 \times 10^{-3}) = 2.79$ (1).	[5]
Q27	Q27: 27. A* Challenge: Weak Acid K_a , $\text{p}K_a$ & Pure Water K_w Temperature Effects 27 27. (a) Approx 1: $[\text{H}^+] = [\text{A}^-]$ (ionization of water is negligible) (1). Approx 2: $[\text{HA}]_{\text{eqm}} \approx [\text{HA}]_{\text{initial}}$ (dissociation is negligible) (1). 27. (b) $K_a = [\text{H}^+]^2 / [\text{HA}] \Rightarrow [\text{H}^+] = \sqrt{K_a \times [\text{HA}]} = \sqrt{1.74 \times 10^{-5} \times 0.150} = \sqrt{2.61 \times 10^{-6}} = 1.615 \times 10^{-3} \text{ mol dm}^{-3}$ (2). $\text{pH} = -\log(1.615 \times 10^{-3}) = 2.79$ (1).	[5]
Q28	Q28: 28. A* Challenge: Weak Acid K_a , $\text{p}K_a$ & Pure Water K_w Temperature Effects 28 28. (a) Approx 1: $[\text{H}^+] = [\text{A}^-]$ (ionization of water is negligible) (1). Approx 2: $[\text{HA}]_{\text{eqm}} \approx [\text{HA}]_{\text{initial}}$ (dissociation is negligible) (1). 28. (b) $K_a = [\text{H}^+]^2 / [\text{HA}] \Rightarrow [\text{H}^+] = \sqrt{K_a \times [\text{HA}]} = \sqrt{1.74 \times 10^{-5} \times 0.150} = \sqrt{2.61 \times 10^{-6}} = 1.615 \times 10^{-3} \text{ mol dm}^{-3}$ (2). $\text{pH} = -\log(1.615 \times 10^{-3}) = 2.79$ (1).	[5]
Q29	Q29: 29. A* Challenge: Weak Acid K_a , $\text{p}K_a$ & Pure Water K_w Temperature Effects 29 29. (a) Approx 1: $[\text{H}^+] = [\text{A}^-]$ (ionization of water is negligible) (1). Approx 2: $[\text{HA}]_{\text{eqm}} \approx [\text{HA}]_{\text{initial}}$ (dissociation is negligible) (1). 29. (b) $K_a = [\text{H}^+]^2 / [\text{HA}] \Rightarrow [\text{H}^+] = \sqrt{K_a \times [\text{HA}]} = \sqrt{1.74 \times 10^{-5} \times 0.150} = \sqrt{2.61 \times 10^{-6}} = 1.615 \times 10^{-3} \text{ mol dm}^{-3}$ (2). $\text{pH} = -\log(1.615 \times 10^{-3}) = 2.79$ (1).	[5]

Q30	Q30: 30. A* Challenge: Weak Acid Ka, pKa & Pure Water Kw Temperature Effects 30 30. (a) Approx 1: $[H^+] = [A^-]$ (ionization of water is negligible) (1). Approx 2: $[HA]_{eqm} \approx [HA]_{initial}$ (dissociation is negligible) (1). 30. (b) $K_a = \frac{[H^+][A^-]}{[HA]} \Rightarrow [H^+] = \sqrt{K_a \times [HA]} = \sqrt{1.74 \times 10^{-5} \times 0.150} = \sqrt{2.61 \times 10^{-6}} = 1.615 \times 10^{-3} \text{ mol dm}^{-3}$ (2). $pH = -\log(1.615 \times 10^{-3}) = 2.79$ (1).	[5]
Q31	Q31: 31. A* Challenge: Weak Acid Ka, pKa & Pure Water Kw Temperature Effects 31 31. (a) Approx 1: $[H^+] = [A^-]$ (ionization of water is negligible) (1). Approx 2: $[HA]_{eqm} \approx [HA]_{initial}$ (dissociation is negligible) (1). 31. (b) $K_a = \frac{[H^+][A^-]}{[HA]} \Rightarrow [H^+] = \sqrt{K_a \times [HA]} = \sqrt{1.74 \times 10^{-5} \times 0.150} = \sqrt{2.61 \times 10^{-6}} = 1.615 \times 10^{-3} \text{ mol dm}^{-3}$ (2). $pH = -\log(1.615 \times 10^{-3}) = 2.79$ (1).	[5]
Q32	Q32: 32. A* Challenge: Weak Acid Ka, pKa & Pure Water Kw Temperature Effects 32 32. (a) Approx 1: $[H^+] = [A^-]$ (ionization of water is negligible) (1). Approx 2: $[HA]_{eqm} \approx [HA]_{initial}$ (dissociation is negligible) (1). 32. (b) $K_a = \frac{[H^+][A^-]}{[HA]} \Rightarrow [H^+] = \sqrt{K_a \times [HA]} = \sqrt{1.74 \times 10^{-5} \times 0.150} = \sqrt{2.61 \times 10^{-6}} = 1.615 \times 10^{-3} \text{ mol dm}^{-3}$ (2). $pH = -\log(1.615 \times 10^{-3}) = 2.79$ (1).	[5]
Q33	Q33: 33. A* Challenge: Weak Acid Ka, pKa & Pure Water Kw Temperature Effects 33 33. (a) Approx 1: $[H^+] = [A^-]$ (ionization of water is negligible) (1). Approx 2: $[HA]_{eqm} \approx [HA]_{initial}$ (dissociation is negligible) (1). 33. (b) $K_a = \frac{[H^+][A^-]}{[HA]} \Rightarrow [H^+] = \sqrt{K_a \times [HA]} = \sqrt{1.74 \times 10^{-5} \times 0.150} = \sqrt{2.61 \times 10^{-6}} = 1.615 \times 10^{-3} \text{ mol dm}^{-3}$ (2). $pH = -\log(1.615 \times 10^{-3}) = 2.79$ (1).	[5]
Q34	Q34: 34. A* Challenge: Weak Acid Ka, pKa & Pure Water Kw Temperature Effects 34 34. (a) Approx 1: $[H^+] = [A^-]$ (ionization of water is negligible) (1). Approx 2: $[HA]_{eqm} \approx [HA]_{initial}$ (dissociation is negligible) (1). 34. (b) $K_a = \frac{[H^+][A^-]}{[HA]} \Rightarrow [H^+] = \sqrt{K_a \times [HA]} = \sqrt{1.74 \times 10^{-5} \times 0.150} = \sqrt{2.61 \times 10^{-6}} = 1.615 \times 10^{-3} \text{ mol dm}^{-3}$ (2). $pH = -\log(1.615 \times 10^{-3}) = 2.79$ (1).	[5]
Q35	Q35: 35. A* Challenge: Weak Acid Ka, pKa & Pure Water Kw Temperature Effects 35 35. (a) Approx 1: $[H^+] = [A^-]$ (ionization of water is negligible) (1). Approx 2: $[HA]_{eqm} \approx [HA]_{initial}$ (dissociation is negligible) (1). 35. (b) $K_a = \frac{[H^+][A^-]}{[HA]} \Rightarrow [H^+] = \sqrt{K_a \times [HA]} = \sqrt{1.74 \times 10^{-5} \times 0.150} = \sqrt{2.61 \times 10^{-6}} = 1.615 \times 10^{-3} \text{ mol dm}^{-3}$ (2). $pH = -\log(1.615 \times 10^{-3}) = 2.79$ (1).	[5]
Q36	Q36: 36. A* Challenge: Weak Acid Ka, pKa & Pure Water Kw Temperature Effects 36 36. (a) Approx 1: $[H^+] = [A^-]$ (ionization of water is negligible) (1). Approx 2: $[HA]_{eqm} \approx [HA]_{initial}$ (dissociation is negligible) (1). 36. (b) $K_a = \frac{[H^+][A^-]}{[HA]} \Rightarrow [H^+] = \sqrt{K_a \times [HA]} = \sqrt{1.74 \times 10^{-5} \times 0.150} = \sqrt{2.61 \times 10^{-6}} = 1.615 \times 10^{-3} \text{ mol dm}^{-3}$ (2). $pH = -\log(1.615 \times 10^{-3}) = 2.79$ (1).	[5]
Q37	Q37: 37. A* Challenge: Weak Acid Ka, pKa & Pure Water Kw Temperature Effects 37 37. (a) Approx 1: $[H^+] = [A^-]$ (ionization of water is negligible) (1). Approx 2: $[HA]_{eqm} \approx [HA]_{initial}$ (dissociation is negligible) (1). 37. (b) $K_a = \frac{[H^+][A^-]}{[HA]} \Rightarrow [H^+] = \sqrt{K_a \times [HA]} = \sqrt{1.74 \times 10^{-5} \times 0.150} = \sqrt{2.61 \times 10^{-6}} = 1.615 \times 10^{-3} \text{ mol dm}^{-3}$ (2). $pH = -\log(1.615 \times 10^{-3}) = 2.79$ (1).	[5]
Q38	Q38: 38. A* Challenge: Weak Acid Ka, pKa & Pure Water Kw Temperature Effects 38 38. (a) Approx 1: $[H^+] = [A^-]$ (ionization of water is negligible) (1). Approx 2: $[HA]_{eqm} \approx [HA]_{initial}$ (dissociation is negligible) (1). 38. (b) $K_a = \frac{[H^+][A^-]}{[HA]} \Rightarrow [H^+] = \sqrt{K_a \times [HA]} = \sqrt{1.74 \times 10^{-5} \times 0.150} = \sqrt{2.61 \times 10^{-6}} = 1.615 \times 10^{-3} \text{ mol dm}^{-3}$ (2). $pH = -\log(1.615 \times 10^{-3}) = 2.79$ (1).	[5]
Q39	Q39: 39. A* Challenge: Weak Acid Ka, pKa & Pure Water Kw Temperature Effects 39 39. (a) Approx 1: $[H^+] = [A^-]$ (ionization of water is negligible) (1). Approx 2: $[HA]_{eqm} \approx [HA]_{initial}$ (dissociation is negligible) (1). 39. (b) $K_a = \frac{[H^+][A^-]}{[HA]} \Rightarrow [H^+] = \sqrt{K_a \times [HA]} = \sqrt{1.74 \times 10^{-5} \times 0.150} = \sqrt{2.61 \times 10^{-6}} = 1.615 \times 10^{-3} \text{ mol dm}^{-3}$ (2). $pH = -\log(1.615 \times 10^{-3}) = 2.79$ (1).	[5]
Q40	Q40: 40. A* Challenge: Weak Acid Ka, pKa & Pure Water Kw Temperature Effects 40 40. (a) Approx 1: $[H^+] = [A^-]$ (ionization of water is negligible) (1). Approx 2: $[HA]_{eqm} \approx [HA]_{initial}$ (dissociation is negligible) (1). 40. (b) $K_a = \frac{[H^+][A^-]}{[HA]} \Rightarrow [H^+] = \sqrt{K_a \times [HA]} = \sqrt{1.74 \times 10^{-5} \times 0.150} = \sqrt{2.61 \times 10^{-6}} = 1.615 \times 10^{-3} \text{ mol dm}^{-3}$ (2). $pH = -\log(1.615 \times 10^{-3}) = 2.79$ (1).	[5]

Q41	Q41: 41. A* Challenge: Weak Acid Ka, pKa & Pure Water Kw Temperature Effects 41 41. (a) Approx 1: $[H^+] = [A^-]$ (ionization of water is negligible) (1). Approx 2: $[HA]_{eqm} \approx [HA]_{initial}$ (dissociation is negligible) (1). 41. (b) $K_a = \frac{[H^+][A^-]}{[HA]} \Rightarrow [H^+] = \sqrt{K_a \times [HA]} = \sqrt{1.74 \times 10^{-5} \times 0.150} = \sqrt{2.61 \times 10^{-6}} = 1.615 \times 10^{-3} \text{ mol dm}^{-3}$ (2). $pH = -\log(1.615 \times 10^{-3}) = 2.79$ (1).	[5]
Q42	Q42: 42. A* Challenge: Weak Acid Ka, pKa & Pure Water Kw Temperature Effects 42 42. (a) Approx 1: $[H^+] = [A^-]$ (ionization of water is negligible) (1). Approx 2: $[HA]_{eqm} \approx [HA]_{initial}$ (dissociation is negligible) (1). 42. (b) $K_a = \frac{[H^+][A^-]}{[HA]} \Rightarrow [H^+] = \sqrt{K_a \times [HA]} = \sqrt{1.74 \times 10^{-5} \times 0.150} = \sqrt{2.61 \times 10^{-6}} = 1.615 \times 10^{-3} \text{ mol dm}^{-3}$ (2). $pH = -\log(1.615 \times 10^{-3}) = 2.79$ (1).	[5]
Q43	Q43: 43. A* Challenge: Weak Acid Ka, pKa & Pure Water Kw Temperature Effects 43 43. (a) Approx 1: $[H^+] = [A^-]$ (ionization of water is negligible) (1). Approx 2: $[HA]_{eqm} \approx [HA]_{initial}$ (dissociation is negligible) (1). 43. (b) $K_a = \frac{[H^+][A^-]}{[HA]} \Rightarrow [H^+] = \sqrt{K_a \times [HA]} = \sqrt{1.74 \times 10^{-5} \times 0.150} = \sqrt{2.61 \times 10^{-6}} = 1.615 \times 10^{-3} \text{ mol dm}^{-3}$ (2). $pH = -\log(1.615 \times 10^{-3}) = 2.79$ (1).	[5]
Q44	Q44: 44. A* Challenge: Weak Acid Ka, pKa & Pure Water Kw Temperature Effects 44 44. (a) Approx 1: $[H^+] = [A^-]$ (ionization of water is negligible) (1). Approx 2: $[HA]_{eqm} \approx [HA]_{initial}$ (dissociation is negligible) (1). 44. (b) $K_a = \frac{[H^+][A^-]}{[HA]} \Rightarrow [H^+] = \sqrt{K_a \times [HA]} = \sqrt{1.74 \times 10^{-5} \times 0.150} = \sqrt{2.61 \times 10^{-6}} = 1.615 \times 10^{-3} \text{ mol dm}^{-3}$ (2). $pH = -\log(1.615 \times 10^{-3}) = 2.79$ (1).	[5]
Q45	Q45: 45. A* Challenge: Weak Acid Ka, pKa & Pure Water Kw Temperature Effects 45 45. (a) Approx 1: $[H^+] = [A^-]$ (ionization of water is negligible) (1). Approx 2: $[HA]_{eqm} \approx [HA]_{initial}$ (dissociation is negligible) (1). 45. (b) $K_a = \frac{[H^+][A^-]}{[HA]} \Rightarrow [H^+] = \sqrt{K_a \times [HA]} = \sqrt{1.74 \times 10^{-5} \times 0.150} = \sqrt{2.61 \times 10^{-6}} = 1.615 \times 10^{-3} \text{ mol dm}^{-3}$ (2). $pH = -\log(1.615 \times 10^{-3}) = 2.79$ (1).	[5]
Q46	Q46: 46. A* Challenge: Weak Acid Ka, pKa & Pure Water Kw Temperature Effects 46 46. (a) Approx 1: $[H^+] = [A^-]$ (ionization of water is negligible) (1). Approx 2: $[HA]_{eqm} \approx [HA]_{initial}$ (dissociation is negligible) (1). 46. (b) $K_a = \frac{[H^+][A^-]}{[HA]} \Rightarrow [H^+] = \sqrt{K_a \times [HA]} = \sqrt{1.74 \times 10^{-5} \times 0.150} = \sqrt{2.61 \times 10^{-6}} = 1.615 \times 10^{-3} \text{ mol dm}^{-3}$ (2). $pH = -\log(1.615 \times 10^{-3}) = 2.79$ (1).	[5]
Q47	Q47: 47. A* Challenge: Weak Acid Ka, pKa & Pure Water Kw Temperature Effects 47 47. (a) Approx 1: $[H^+] = [A^-]$ (ionization of water is negligible) (1). Approx 2: $[HA]_{eqm} \approx [HA]_{initial}$ (dissociation is negligible) (1). 47. (b) $K_a = \frac{[H^+][A^-]}{[HA]} \Rightarrow [H^+] = \sqrt{K_a \times [HA]} = \sqrt{1.74 \times 10^{-5} \times 0.150} = \sqrt{2.61 \times 10^{-6}} = 1.615 \times 10^{-3} \text{ mol dm}^{-3}$ (2). $pH = -\log(1.615 \times 10^{-3}) = 2.79$ (1).	[5]
Q48	Q48: 48. A* Challenge: Weak Acid Ka, pKa & Pure Water Kw Temperature Effects 48 48. (a) Approx 1: $[H^+] = [A^-]$ (ionization of water is negligible) (1). Approx 2: $[HA]_{eqm} \approx [HA]_{initial}$ (dissociation is negligible) (1). 48. (b) $K_a = \frac{[H^+][A^-]}{[HA]} \Rightarrow [H^+] = \sqrt{K_a \times [HA]} = \sqrt{1.74 \times 10^{-5} \times 0.150} = \sqrt{2.61 \times 10^{-6}} = 1.615 \times 10^{-3} \text{ mol dm}^{-3}$ (2). $pH = -\log(1.615 \times 10^{-3}) = 2.79$ (1).	[5]
Q49	Q49: 49. A* Challenge: Weak Acid Ka, pKa & Pure Water Kw Temperature Effects 49 49. (a) Approx 1: $[H^+] = [A^-]$ (ionization of water is negligible) (1). Approx 2: $[HA]_{eqm} \approx [HA]_{initial}$ (dissociation is negligible) (1). 49. (b) $K_a = \frac{[H^+][A^-]}{[HA]} \Rightarrow [H^+] = \sqrt{K_a \times [HA]} = \sqrt{1.74 \times 10^{-5} \times 0.150} = \sqrt{2.61 \times 10^{-6}} = 1.615 \times 10^{-3} \text{ mol dm}^{-3}$ (2). $pH = -\log(1.615 \times 10^{-3}) = 2.79$ (1).	[5]
Q50	Q50: 50. A* Challenge: Weak Acid Ka, pKa & Pure Water Kw Temperature Effects 50 50. (a) Approx 1: $[H^+] = [A^-]$ (ionization of water is negligible) (1). Approx 2: $[HA]_{eqm} \approx [HA]_{initial}$ (dissociation is negligible) (1). 50. (b) $K_a = \frac{[H^+][A^-]}{[HA]} \Rightarrow [H^+] = \sqrt{K_a \times [HA]} = \sqrt{1.74 \times 10^{-5} \times 0.150} = \sqrt{2.61 \times 10^{-6}} = 1.615 \times 10^{-3} \text{ mol dm}^{-3}$ (2). $pH = -\log(1.615 \times 10^{-3}) = 2.79$ (1).	[5]

SECTION D: EMPIRICAL EXAMINER FAQs & COMMON TRAPS (10 FAQs) — 14A

FAQ #1: pH Decimal Places Rule

Category: Significant Figures

COMMON EXAMINER TRAP / MISTAKE:

Writing pH to 1 decimal place or rounding incorrectly.

MODEL ANSWER / EXAMINER EXPECTATION:

pH decimal places represent significant figures. pH should ALWAYS be given to 2 DECIMAL PLACES in Edexcel exams (e.g. 2.79, not 2.8).

FAQ #2: Strong Dibasic Acid pH Calculation

Category: Stoichiometric Factor

COMMON EXAMINER TRAP / MISTAKE:

Assuming $[H^+] = [\text{acid}]$ for H_2SO_4 .

MODEL ANSWER / EXAMINER EXPECTATION:

Sulfuric acid H_2SO_4 is dibasic: 1 mole H_2SO_4 produces 2 moles of H^+ upon complete dissociation $\rightarrow [H^+] = 2 \times [H_2SO_4]$.

FAQ #3: Pure Water pH at Temperatures Other Than 298K

Category: Kw Temperature Effect

COMMON EXAMINER TRAP / MISTAKE:

Assuming pure water is always pH 7.00 at all temperatures.

MODEL ANSWER / EXAMINER EXPECTATION:

Water dissociation $H_2O \rightleftharpoons H^+ + OH^-$ is ENDOTHERMIC. Increasing temperature increases $K_w \rightarrow [H^+]$ increases $\rightarrow$ pH of pure water DECREASES below 7.00. However, water remains NEUTRAL because $[H^+] = [OH^-]$.

FAQ #4: Definition of Neutral Solution

Category: pH Definitions

COMMON EXAMINER TRAP / MISTAKE:

Defining a neutral solution as pH = 7.

MODEL ANSWER / EXAMINER EXPECTATION:

A neutral solution is defined as $[H^+] = [OH^-]$. It is ONLY pH 7.00 at 298 K where $K_w = 1.0 \times 10^{-14}$.

FAQ #5: Two Approximations in Weak Acid pH Calculations

Category: Ka Assumptions

COMMON EXAMINER TRAP / MISTAKE:

Forgetting the two key assumptions when using $[H^+] = \sqrt{K_a \times [HA]}$.

MODEL ANSWER / EXAMINER EXPECTATION:

Assumption 1: $[H^+] = [A^-]$ (ignores H^+ from water). Assumption 2: $[HA]_{eqm} = [HA]_{start}$ (ignores amount of HA dissociated). Both assumptions fail for very dilute or stronger weak acids.

FAQ #6: pKa and Acid Strength Relationship

Category: pKa Values

COMMON EXAMINER TRAP / MISTAKE:

Assuming a larger pKa value means a stronger acid.

MODEL ANSWER / EXAMINER EXPECTATION:

$pK_a = -\log K_a$. A SMALLER pK_a (or larger K_a) means a STRONGER acid. E.g. pK_a 3.75 (methanoic) is stronger than pK_a 4.76 (ethanoic).

FAQ #7: Conjugate Base Strength Relationship**Category: Conjugate Pairs****COMMON EXAMINER TRAP / MISTAKE:**

Stating a strong acid has a strong conjugate base.

MODEL ANSWER / EXAMINER EXPECTATION:

The stronger the acid, the WEAKER its conjugate base. E.g. HCl is a very strong acid, so Cl^- is an extremely weak conjugate base with negligible proton affinity.

FAQ #8: Dilution of Weak Acid Effect on pH**Category: Ostwald Dilution****COMMON EXAMINER TRAP / MISTAKE:**

Assuming diluting a weak acid by 10x increases pH by 1.0 unit.

MODEL ANSWER / EXAMINER EXPECTATION:

Diluting a strong acid 10x increases pH by 1.0 unit. Diluting a WEAK acid 10x increases pH by approximately 0.5 units because dilution shifts equilibrium $HA \rightleftharpoons H^+ + A^-$ to the right (increases % dissociation).

FAQ #9: Kw Units**Category: Kw Expression****COMMON EXAMINER TRAP / MISTAKE:**

Writing K_w units as mol dm^{-3} .

MODEL ANSWER / EXAMINER EXPECTATION:

$K_w = [H^+][OH^-]$. The units of K_w are $(\text{mol dm}^{-3})(\text{mol dm}^{-3}) = \text{mol}^2 \text{dm}^{-6}$.

FAQ #10: Base Hydroxide Concentration Calculation**Category: Base Calculations****COMMON EXAMINER TRAP / MISTAKE:**

Calculating pH directly from $[OH^-]$ using $-\log[OH^-]$.

MODEL ANSWER / EXAMINER EXPECTATION:

$-\log[OH^-]$ gives pOH, NOT pH! To find pH of a base: first find $[H^+] = K_w / [OH^-]$, then calculate $pH = -\log[H^+]$. Alternatively $pH = 14 - pOH$ at 298 K.